

When Sample Selection Bias Precipitates Model Collapse

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Theory Claims

• **Proposition 1: Replace paradigm**

Under the *Replace* paradigm, as $t \rightarrow \infty$,

$$\Sigma_t \xrightarrow{a.s.} \mathbf{0}, \quad \mathbb{E}[\mathbb{W}_2^2(\mathcal{N}_t, \mathcal{N}_0)] \rightarrow \infty.$$

• **Proposition 2: Accumulate paradigm**

Under unbiased accumulation,

$$\mathbb{E}[(\bar{\mu}_t - \bar{\mu}_0)^2] \rightarrow (1 - \alpha_n) \bar{\Sigma}_0, \quad \mathbb{E}[\bar{\Sigma}_t] \rightarrow \alpha_n \bar{\Sigma}_0.$$

Thus $\mathbb{E}[\mathbb{W}_2^2] \rightarrow 2(1 - \sqrt{\alpha_n}) \text{Tr}(\bar{\Sigma}_0)$.

• **Theorem 1: Selection bias precipitates collapse**

With top- α selection toward a local ideal $\mathbf{u}^*$,

$$\|\bar{\mu}_t - \mathbf{u}^*\|^2 \rightarrow 0, \quad \bar{\Sigma}_t \rightarrow \mathbf{0}.$$

Local fidelity can erase global diversity.

• **Theorem 2: Collapse has an explicit rate**

If the trajectory remains in the local basin,

$$\text{Tr}(\bar{\Sigma}_t) = \mathcal{O}_{a.s.}(t^{-\psi}).$$

The tail erosion follows a power-law decay.

• **Theorem 3: Collapse costs true-manifold risk**

For filtered distribution $\mathcal{D}_t$ and true manifold $\mathcal{D}^*$,

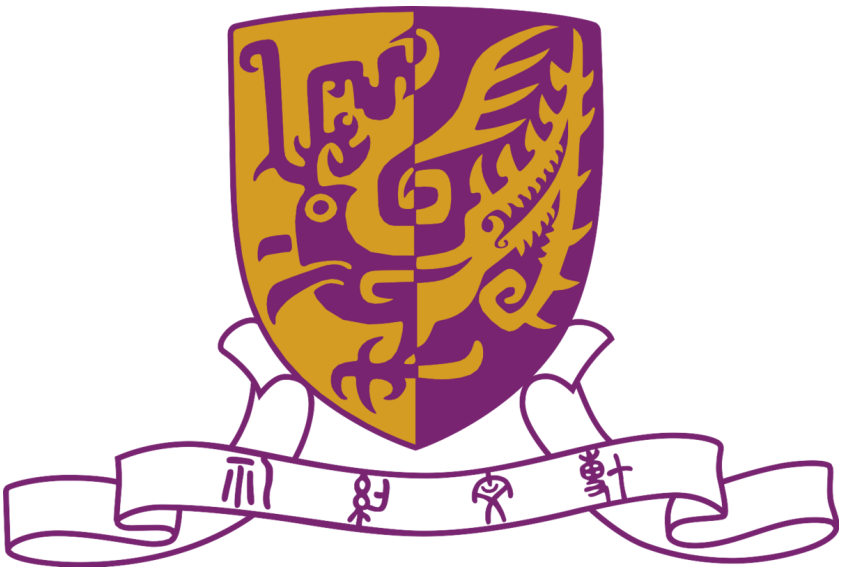
$$\mathcal{R}_{\mathcal{D}^*}(h_t; g^*) \leq 2\ell\epsilon \mathbb{W}_p(\mathcal{D}_t, \mathcal{D}^*) + \mathcal{R}_{\mathcal{D}_t}(h_t; g_t) + \mathcal{O}(\ell\delta).$$

Proxy Principle

A collaborative Wasserstein barycenter broadens the verifier:

$$\mathcal{Q}^* = \arg \min_{\mathcal{Q}} \sum_{k=1}^K \lambda_k \mathbb{W}_p^p(\mathcal{Q}, \mathcal{Q}_k).$$

Selection is protective only when this proxy covers the modes
the loop must preserve.



Scan

OpenReview paper
and supplement.

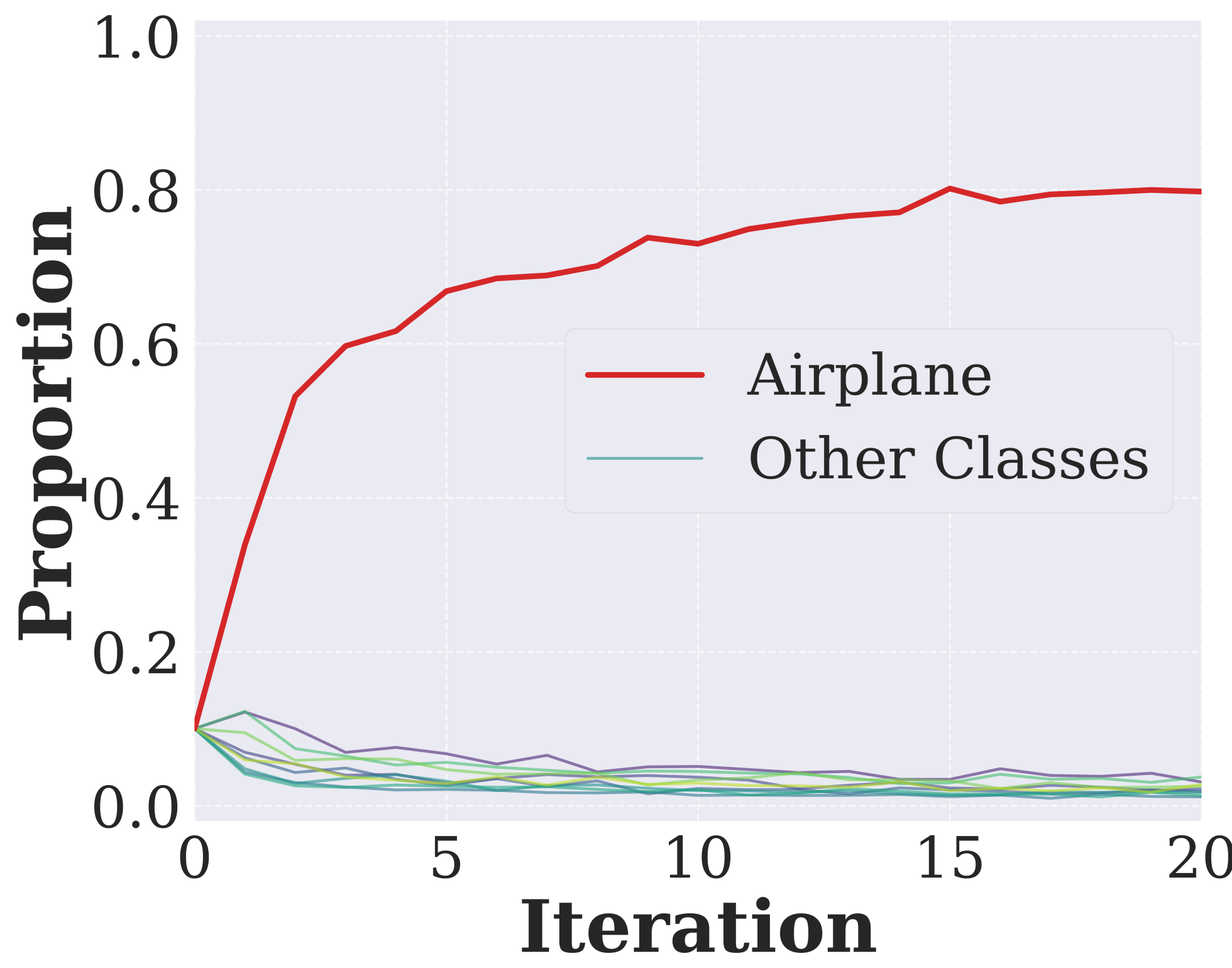
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Bias prunes tails. Silos collapse faster. Wasserstein proxies restore tail coverage. No raw data sharing.

Experimental Takeaway

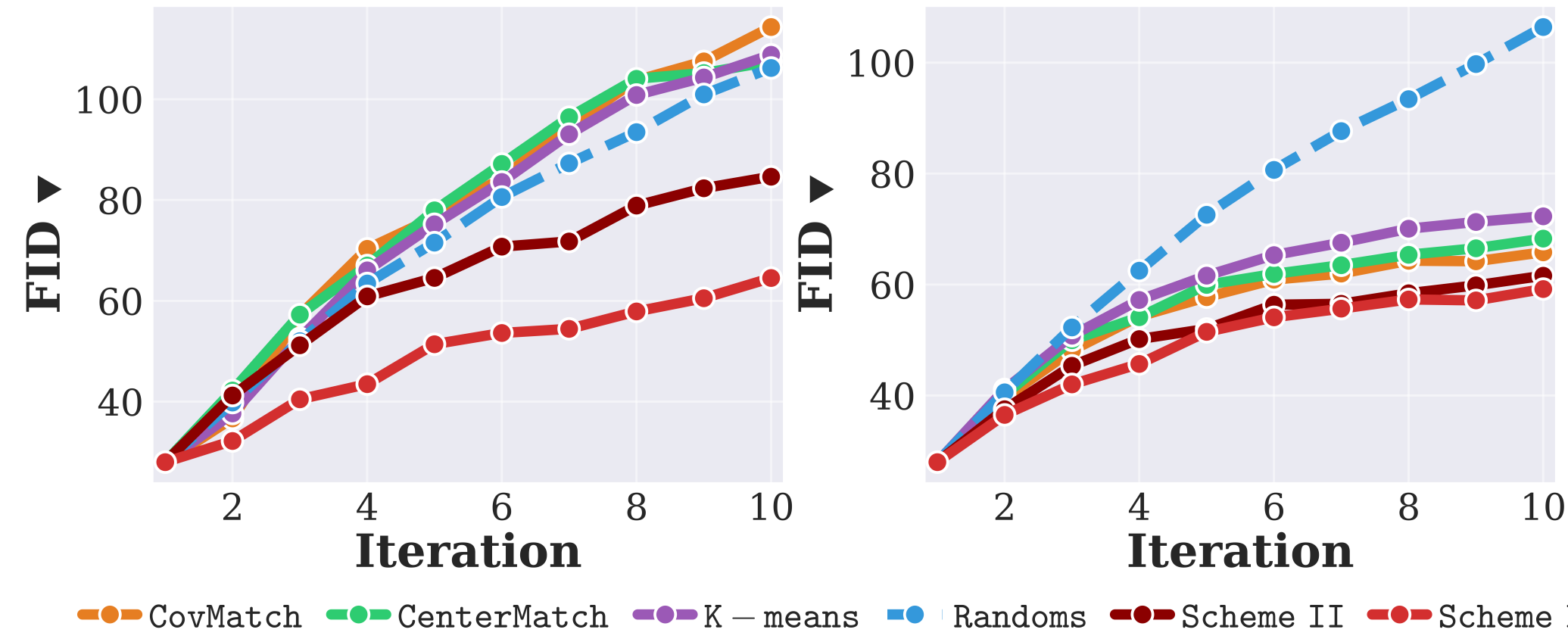
- Airplane-only reference homogenizes CIFAR-10.
- Scheme I improves CIFAR-10 FID: 106 to 71.
- Collaborative proxies keep tail modes visible.

Collapse Signal



One-class verification pulls the sample loop toward Airplane
while the other classes become tail modes.

FID Trend



On non-IID references, collaborative proxy selection lowers FID
relative to random selection and common matching baselines.

